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LIBRARY DESIGN: ITS IMPLICATIONS ON LEARNING MOTIVATION

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CHAPTER 1
THE PROBLEM AND ITS SCOPE

Background of the study

In the 21st century, libraries in basic education have transformed from simple book repositories into dynamic learning environments that influence students' mental, emotional, and motivational well-being. Modern libraries increasingly resemble "learning commons" with digital technologies, flexible seating, and quiet zones that cater to diverse learning preferences.

Studies highlight the importance of library design in fostering student engagement. Kim (2019) noted that academic libraries in South Korea support both individual study and collaborative learning. Similarly, Foo and Majid (2021) found that Asian libraries with digital hubs and contemporary furniture encourage students to revisit and interact with the space. Elements such as layout, lighting, acoustics, ventilation, and furniture significantly shape how students organize study time and complete academic tasks (Pagalilauan et al., 2022). Adequate lighting, proper airflow, and ergonomic furniture consistently rank among the top factors sustaining student motivation (2019 Survey on Students' Opinion).

Environmental comfort is equally important. Daramola (2021) found that noise control and access to technology increase satisfaction and study duration, while Oyelekan and Ede (2022) noted that combining quiet and collaborative zones enhances engagement. Moreover, biophilic design elements such as natural light, greenery, and sustainable materials positively impact concentration and well-being (Nguyen, 2020; Khoir et al., 2020; Li et al., 2024).

In basic education, school libraries are essential for developing reading habits and literacy. Safitri and Widagdo (2021) found that well-organized libraries significantly influence Grade IV pupils' reading interest. Azhari and Ramadan (2022) emphasized that frequent library visits nurture a positive reading culture, while Figuration and Ormilla (2021) observed improved reading performance through structured library programs. Both traditional and digital libraries offer accessible materials that enhance student engagement (Mulyawati & Marini, 2022).

Despite these benefits, challenges remain in the Philippines. Villanueva (2019) reported that clear layouts and dedicated zones improve student motivation, while Santos and Reyes (2020) highlighted the importance of lighting, cleanliness, and airflow. However, Perez and Diaz (2021) noted poor ventilation and inadequate collections, Gonzales and Buenavista (2019) identified outdated furniture, and Dela Cruz (2019) observed that students often seek alternative study spaces due to limited online access. These issues reduce the appeal and effectiveness of school libraries.

At Saint Michael College of Caraga (SMCC) Basic Education Library, interviews with students revealed concerns about noise disturbances, limited seating, outdated furniture, insufficient learning materials, and inconsistent room temperature. These factors hinder comfort, concentration, and study effectiveness. Recognizing that the physical environment is crucial for motivation and academic persistence, this study aims to examine how specific design elements including layout, lighting, acoustics, ergonomic furniture, technological infrastructure, and natural features affect students' motivation to learn. The goal is to identify which aspects of SMCC's library design most positively

influence learning motivation and to provide recommendations for a more conducive, inspiring study environment.

This research is significant for multiple stakeholders. For students, it informs improvements that enhance focus, motivation, and academic performance. For faculty and staff, it offers insights into how library design supports diverse teaching strategies and learning needs. For SMCC administration, it provides evidence-based guidance for future renovations, furniture upgrades, and resource allocation. Beyond SMCC, the study contributes to the limited research on academic library design in the Philippines, offering practical guidelines for creating student-centered learning spaces.

Related Literature

The library is an essential part of any educational institution. It serves as the heart of learning, where students can access information, study independently, and develop a deeper understanding of their lessons. Beyond being a place for books, the library provides a quiet and supportive environment that encourages focus, creativity, and a love for reading. It also helps students build research skills, explore new ideas, and become lifelong learners. In essence, the library plays a vital role in shaping students.

Library design is the process of planning and setting up the physical, technological, and environmental components of a library to create a space that encourages learning, keeps people interested, and motivates them. Dianing and Acedera (2025) note that library design includes elements such as space layout, lighting, acoustics, ergonomic furniture, technology infrastructure, and green spaces to make users more comfortable, help them focus, and improve their overall academic experience.

Physical Layout and Organization

According to Dianing and Acedera (2025), library design is a multidimensional concept that encompasses several key elements that encourage students to learn and engage in their work. One of these factors is the physical layout and organization, which means how the library spaces are set up to make them easy to get to, comfortable, and valuable. A well-thought-out layout makes it easy for people to move around the space, find what they need, and choose areas that are good for studying alone or working with others, thereby motivating them to learn.

Similarly, Nukulkij et al. (2024) argue that the physical environment in schools is essential to maintaining students' health. Libraries are significant parts of student life that contribute to it. In line with this, Merga (2021) said that students who use the library to "relax and recharge" often associate this feeling with the comfortable furniture and the library's general atmosphere. Likewise, Willis et al. (2019) found that when students discuss their ideal library spaces, they value feeling connected and calm. These features are linked to both physical and mental health because they give students safe places to relax, think, and feel at ease in the school setting.

Furthermore, Han and Wan (2021) added to this by listing different aspects of a space including lighting, airflow, temperature, furniture, and ease of access that affect how students learn, how well they do, and how well they feel overall. If students' study spaces are set up to fit their preferred study methods and learning styles, they are more likely to be used for both individual and group study.

In addition**,** Shi and Liu (2021) said that traditional university libraries need to do more. Technology is becoming more important, and people's needs are changing. These days, libraries are more than just places to read. They also have digital resources, training sessions, academic seminars, and art exhibits to help people learn and get involved. Moreover, Ilett (2019) also noted that more research is needed on how first-generation college students use library spaces, as these students often face unique challenges in college. Supporting this, according to a study by Couture et al. (2021), first-generation students felt confident and like they belonged in libraries, which are welcoming places that help their academic and personal growth. Inclusion also includes students who have

special needs. Accordingly, Natal & Remaklus (2023) emphasized the importance of making library spaces accessible by ensuring physical accessibility, placing artwork in easy-to-reach locations for people with limited mobility and vision, and using legible labels and signs.

Overall, these studies show that a well-thought-out library setting one that encourages comfort, acceptance, adaptability, and ease of access can significantly improve students' motivation, engagement, and health. Libraries that are comfortable for everyone and meet a variety of learning needs are important places for academic success and overall growth.

Lighting and Acoustics

Lighting and sound are two other important factors that Dianing and Acedera (2025) stress. Specifically, Good lighting and sound design help you concentrate and see clearly, and keeping the room quiet and free of distractions is important for studying. These things work together to make a balanced environment that helps people focus and keeps their minds from getting tired, so they can keep working on their projects. In support of this, several new studies have shown that improving lighting design and technology in learning and library spaces makes them more comfortable, functional, and user-friendly. In 2022, Dabaj et al. looked at how different shading devices affected daylight performance and visual comfort in a Rasht reading room. Their research showed that perforated solar screens let in more light, but Venetian blinds were better at blocking out glare and making the room look better overall.

As a result, The study showed how important it is to choose shading systems that work well with the local climate to achieve the right amount of light, save energy, and ensure people are comfortable. Similarly, Redondo et al. (2022) conducted a similar study, using orthogonal arrays and listing methods to identify the optimal parameters for perforated façade designs in Seville offices. Their statistical model correctly predicted façade configurations, increasing daylight autonomy and uniformity while reducing simulation time, underscoring the value of data-driven approaches to sustainable lighting design.

Moreover, Also, Quiles-Rodríguez and Palau (2024) looked into how colored LED lighting affected students' emotional and cognitive learning. Their experiments showed that colored lighting improved focus and emotional engagement. In other words, Suggesting that dynamic lighting systems can make learning environments more engaging and interactive. In the same way, Timothy et al. (2024) examined how daylight performed in an architecture library using different combinations of windows and screen walls. Their findings showed that installing a screen wall 900–1200 mm from a central window, 1100 mm above the floor, increased daylight levels by 36–38%, providing helpful information for designing energy-efficient library lighting.

Supporting these findings, Aly, Kamel, and Soliman (2021) looked at the quality of daylight in university libraries in Egypt. They found that changes such as improved window design and the addition of light shelves significantly increased light levels and reduced glare, improving visual comfort and saving energy. Likewise, In the same way, Ali, Jabeen, and Zahid (2023) examined how advanced technology and effective management affect

the happiness of library users. According to their research, combining modern technologies and effective management techniques increased users' trust in libraries and improved services overall.

Finally, Lestari and Haryanto (2021) examined the lighting in the reading room at Batam Public Library and found that some areas lacked sufficient light. The results of their simulation showed that a balanced mix of natural and artificial lighting would make reading more comfortable while using the least amount of energy. Taken together, All of these studies show how important it is to carefully plan lighting, ensure a good environment, and develop new technologies to create learning and library spaces that are comfortable, efficient, and inspiring.

Ergonomics of Arrangements

They are also essential for keeping people motivated. According to Dianing and Acedera (2025) state that ergonomically designed furniture and well-designed space arrangements make it easier for students to study for extended periods without becoming uncomfortable or straining. A more inviting place to learn is one with comfortable chairs, tables at the right height, and enough space between study areas.

In support of this, Hernández and Pérez (2022) emphasized the importance of ergonomic seating in learning spaces. They said that poorly designed furniture and space layouts make people uncomfortable and hurt their academic performance. To improve comfort and engagement, they suggested adjustable furniture, enough space for movement, and chairs that support good posture.

Similarly, Ukoha and Adebayo (2023) examined ergonomic library spaces and found that the arrangement of furniture and study areas significantly affects students' satisfaction and engagement with their work. Their research showed that well-organized, ergonomically designed spaces make learning easier on the body and help sustain learning.

As for furniture design and space layout, Santos, Oliveira, and Lima (2024) found that the correct desk height, seating, and distance between workstations directly affect users' comfort and productivity. Their study stressed the importance of using anthropometric data in design decisions to prevent muscle and joint strain and to utilize space better. In agreement David Publisher (2024) agreed, stating that visual comfort, good lighting, and clear pathways for movement are all important for ergonomic optimization. They said that poorly organized environments make people less productive and less able to focus, while well-organized environments encourage engagement and better learning outcomes. Moreover, Dávalos Quevedo (2025) suggested using observational data, surveys, postural assessments, and other methods to redesign educational spaces with an ergonomic focus effectively.

In line with this, the Australian Journal of Educational Technology (2022) stated that combining ergonomics with technology infrastructure makes modern learning spaces more engaging and happier for users. All of these results support the idea that ergonomic evaluation should be ongoing and data-driven to ensure that academic spaces remain comfortable and function well over the long term. Overall, the studies examined show that the arrangement's ergonomics is essential for creating learning spaces that work well

and quickly. Ergonomic principles, well-designed furniture, and thoughtful space planning all make users more comfortable and improve their motivation and academic performance. By following ergonomic standards for seating, movement, workstation placement, and technology use, schools can create spaces that are both useful and conducive to deep learning. Because of this, the ergonomics of arrangement is a key factor in making libraries and classrooms healthier, more productive, and better places to learn overall.

Technology Infrastructure

Infrastructure for technology is an important part of designing a sound library. According to Dianing and Acedera (2025), providing students with access to digital tools, high-speed internet, and multimedia resources enhances learning for both individuals and groups. Access to technology not only makes research easier but also motivates students by incorporating new ways of learning that appeal to those who grew up with it.

Additionally, Technology infrastructure is an important part of good library design because it directly affects students' motivation and interest in learning, as well as their overall educational experience. Dianing and Acedera (2025) stated that technology is an important part of library development, but that other factors, such as layout, lighting, and the presence of green spaces, often have a greater impact on student engagement.

In support of this, The Journal of Universal and Public Education in New Scientific Innovations (2024) also stated that strong technological infrastructure improves learning by creating compelling, engaging learning spaces. In line with this idea, the Information Technology Infrastructure Library (ITIL) framework (2023) demonstrated how well-

organized IT management improves library services, making learning easier and giving students better access to learning materials.

Furthermore, Digital technologies have completely changed the way schools work by making learning more flexible, personalized, and engaging. According to a 2022 PMC study, adding technology to learning spaces makes them more flexible and helps each student learn in their own way. In addition, Educause (2021) found that well-designed learning spaces with modern technology encourage students to work together, be creative, and be active participants.

Moreover, The University of Alberta (n.d.) also noted that technology-integrated school libraries can motivate students by making them easier to use and more interested in learning. Similarly, ScienceDirect (2024) stated that technological infrastructure was a key factor in implementing adaptive learning models in higher education, enabling students to have personalized learning plans that meet their needs.

In addition, Using multimedia technologies in library lessons has also been shown to make students much more interested in learning. May (2019) says that multimedia tools can be used by students with different learning styles and help them remember what they have learned. In support of this, Manguil et al. (2024) and Rone et al. (2023) reported that technology infrastructure performs better when paired with librarian support and opportunities to meet new people. In the same way, Rodrigues and Mandrekar (2020) and Dupa (2019) said that technology is important. However, it needs to be paired with a helpful person and a good physical environment to keep students motivated.

Overall, It is important for modern education that libraries have technology infrastructure, but it only works well when it works with well-designed physical spaces and services that focus on people. When put together, these factors create an environment for learning that encourages students to be interested, work together, be happy, and stay motivated (Manguil et al., 2024).

Green Spaces / Nature Integration

Adding green spaces and nature to library spaces can help people feel better. According to Dianing and Acedera (2025) found that adding natural elements like indoor plants, natural lighting, and views of the outdoors can help people relax, feel less stressed, and think more clearly. These biophilic design elements turn libraries into inspiring and relaxing places that help students stay motivated, on task, and interested.

In support of this, It has been shown that adding green and biophilic elements to libraries increases people's desire to learn. Boyd (2022) and Browning and Rigolon (2019) argue that allowing students to be close to nature on school grounds improves their mental health and academic performance. Additionally, DeLauer et al. (2022) also say that biophilic design makes learning environments that are helpful and caring.

Moreover, Ho, Chiu, and Ho (2022), Demirtas Dogan and Gurpinar (2023), and Fedorowicz-Kruszewska (2022) also stated that green libraries not only raise awareness of environmental issues but also make learning spaces more engaging. Mason et al. (2022) and Noor Ashraf Awad (2022) argue that students can be more motivated to learn if they spend time in green spaces, which can also improve their mood and cognitive performance. Likewise, Peters et al. (2020) also discussed how biophilic design can make

learning and healing spaces more effective.

Furthermore, User experience and engaging the senses are also significant. Siham (2023) and To and Grierson (2024) say that multisensory and nature-inspired designs help students focus, be creative, and enjoy their learning spaces. Similarly, Many researchers, including Vleet et al. (2025) and Mwanzu et al. (2023), say that gardens and courtyards can improve students' mood and motivation.

However, Nagaiah et al. (2024) noted that funding and maintenance challenges can make it difficult to open green libraries. In addition, Lotfy et al. (2022) also stated that libraries need a mix of formal study areas and casual learning areas to meet the needs of all student types. Overall, Green practices are being slowly put in place in libraries, which is both a good sign and a problem. Vrana and Zečević (2020) note that Croatian public libraries are gradually introducing green initiatives. Showing that sustainability is possible, but needs careful planning. Similarly, Zheng et al. (2024) also discussed how library spaces affect how students interact with one another and how engaged they are in learning. They stressed the importance of user-centered planning and space design for motivating students.

All of these studies show that adding green and biophilic elements to library design directly improves students' motivation to learn, engagement, and health. Libraries can be more than places to study when they have flexible spaces, natural elements, and student-centered approaches. These make libraries places that encourage and support learning.

Learning Motivation

Learning motivation is influenced by multiple psychological and environmental

factors that determine a student's willingness to engage in academic tasks. According to the study by Li et al. (2021), learning motivation is shaped by intrinsic and extrinsic drives: intrinsic motivation encourages learners to seek knowledge for personal fulfillment, whereas external rewards or academic outcomes drive extrinsic motivation.

Furthermore, The study also highlights self-efficacy in information literacy as a significant factor, indicating that students who feel confident in locating and using information are more motivated to participate in learning activities. In addition, the perceived usefulness of resources contributes to learning motivation, as students are more engaged when they believe that the available materials support their academic goals.

Moreover, Engagement and learning satisfaction further reinforce motivation by creating positive learning experiences that sustain interest and persistence. Therefore, These factors collectively demonstrate that learning motivation is a multifaceted construct influenced by internal beliefs and perceptions related to resource availability and academic success.

Intrinsic motivation

Indeed, Li et al. (2021) agreed with this, stating that intrinsic motivation leads to more active thinking and a strong desire to learn independently. Similarly, Zhao and Wang (2025) also found that students are more engaged and perform better in school when intrinsically motivated. This is because intrinsically motivated learning activities align with their personal values and goals. Based on field data, Maheshwari (2021) discovered that intrinsic motivation has a significant effect on perceived utility. Specifically, These

variables concern Vietnamese students' willingness to learn and use technology online. Hence, intrinsic drive is seen as a good way to encourage people to make decisions independently in a library makerspace learning setting. However, Cherry (2019) says that having intrinsic motivation does not mean a person will not look for incentives. In this case, it just means that outside incentives are not enough to keep someone motivated. For example, One example of a student who is intrinsically motivated is someone who wants to get a good grade on an assignment. Yet, if the student does not care about the assignment, the chance of getting a good grade will not encourage them to work on the project (Li, 2021).

Overall, Intrinsic motivation means that students engage in purposeful activities because they enjoy them and feel good about their own abilities, rather than to get rewards from external sources (Ryan et al., 2021). Consequently, Students who use the school library to learn and do research are naturally driven to do so because it helps them understand things better, be more creative, and keep going even when things get hard. In this way, This drives people to keep learning throughout their lives and strengthens the emotional connection they have with their education.

Extrinsic Motivation

Extrinsic motivation means doing schoolwork to earn rewards from outside sources, such as grades, praise, or social approval. According to Li et al. (2021), external motivators, such as institutional expectations and assessment requirements, significantly influence students' behavior in school. Moreover, Hussain (2021) also noted that library use typically increases when schoolwork explicitly requires it, suggesting that external

factors encourage people to use learning resources. Similarly, Zhang and Yang (2020) found that students are more likely to use academic tools such as libraries and online platforms when these tools are directly related to their performance or success metrics.

Furthermore, Li et al. (2021) said that extrinsic motivation is just as important as intrinsic motivation because it drives behavior through external rewards and academic expectations. In contrast, Intrinsic motivation drives people to do academic tasks for personal satisfaction. Thus, Students often use external factors to drive their behavior in school, such as grades, praise, meeting course requirements, and following the school's rules. This is especially true when students use libraries, where they must find resources not only for their own pleasure but also to complete assignments, pass tests, or meet the requirements set by their teachers and academic programs. Therefore, extrinsic motivation is a very important factor in how students learn and, in their likelihood, to use academic support systems, such as libraries.

Recent studies also show that external factors beyond academic requirements significantly affect how and why students use libraries. For instance, According to Zhang and Yang (2020), students are more interested in learning activities when they believe that using institutional resources, such as libraries, can directly help them perform better in school. Similarly, Researchers found that students were more likely to visit the library when they knew that using its digital databases, research materials, and learning technologies would help them perform better in school and in their classes. In this way, The library serves as an incentive for students when they believe using it will lead to external benefits, such as good grades or academic praise.

In the same way, Hussain (2021) noted that academic libraries foster intrinsic motivation by serving as important academic support hubs that provide students with access to required readings, course-related resources, and structured study spaces. Hence, Students use library services not just because they are interested, but also because they need to in order to meet academic standards and meet institutional requirements. In line with this idea, Alasmari and Zhang (2019) found that students are more likely to use learning technologies and academic tools available in libraries and other institutional settings when they know these tools will help them do well in school. Overall, These results show that external motivation is a significant factor in library use. Indeed, Students feel obligated to use libraries as part of their academic duties and to meet performance expectations set by others.

Therefore, Overall, these studies show that external motivation significantly affects how students learn, making the library an important outside force that motivates students to do well in school by setting expectations, requiring performance, and pursuing desired academic outcomes.

Self-Efficacy Information Literacy

In terms of information literacy, self-efficacy refers to students' confidence in their ability to find, evaluate, and use information effectively. According to Li et al. (2021), students with higher self-efficacy are more motivated to learn because they believe they can better handle academic challenges.

Similarly, the author Liu (2023) stated that students perform better in school and in research when they are confident in their information literacy skills. Moreover, The

research showed that students who learned to use information effectively became more motivated and more confident in their abilities, thus making it easier for them to find and use information. In addition, Medaille et al. (2021) also looked at how honors students' confidence in their ability to use information changed as they worked on their theses. The results indicated that people who felt more confident in their ability to find and evaluate information were able to finish their research faster.

Furthermore, Chow and Wong (2020) also talked about how important it is to boost students' academic motivation and sense of self-efficacy. In fact, Their research showed that teaching students how to use information effectively not only helps them learn new skills but also boosts their confidence in using those skills in different academic settings. In a related study, the British Journal of Multidisciplinary and Advanced Studies (2023) found that students who are confident in their ability to use information effectively have stronger research skills and are better prepared to address challenges in school.

Overall, These studies show that students' belief in their ability to learn information literacy has a significant effect on their motivation to learn and on their behavior during learning. Therefore, Students will likely do well in school if they believe in their ability to find and use information. Consequently, This helps achieve the study's objective of understanding how a library's layout affects students' drive and belief in their ability to learn.

Perceived Usefulness of Resources

Regarding perceived usefulness, it is how students think that educational facilities and materials help them do well in school. Indeed, It is one of the most important factors

in determining motivation, as Li et al. (2021) noted that learners are more likely to use resources when they believe they will help them reach their goals. Moreover, how useful students perceive library materials to be is closely related to their satisfaction with them. For instance, According to Oyeleye et al. (2021), students in Nigerian universities were more interested and satisfied when library materials were relevant, up to date, and easy to find. Therefore, The results show that how functional library materials are has a direct effect on how students use them to learn.

In the same way, Bankole et al. (2024) found that students' study habits and attitudes toward learning were significantly affected by the ease of access to resources and their perceived usefulness. Thus, When students see resources as valuable and important, they are more motivated to use them and stick to regular study schedules, showing how important resource usefulness is for keeping students motivated to learn.

In addition, In the digital world, Bhaskar et al. (2024) found that users were happier and more likely to continue using digital library platforms when they perceived them as applicable. Specifically, People who thought the digital resources were helpful were more likely to use them regularly and learn more. This illustrates that how helpful someone thinks something is is important in both physical and digital libraries.

Furthermore, Scoulas and De Groote (2024) also reported that faculty members used library resources because they found them helpful, which in turn improved research productivity and overall happiness. Finally, Tang and Tseng (2023) found that when librarians promoted Open Educational Resources (OER), users perceived the resources as more useful, thereby leading to greater engagement and satisfaction. Overall, All of these

studies show that how helpful people think library resources are is a significant factor in their intrinsic motivation, satisfaction, and long-term use of both physical and digital library services.

Student Engagement

According to Li et al. (2021), engagement makes learning more fun and engaging because it makes lessons more active and meaningful. Moreover, New research by MacMillan and Edwards (2025) also suggests that library design features, such as collaborative zones and technology-equipped areas, help people stay engaged, thereby increasing motivation and achievement.

Indeed, Student engagement has become a significant topic of study in education because it directly affects how well students learn, how well they do in school, and how happy they are with their lives in general. For example, Trowler (2021) says that student engagement includes the time, effort, and mental investment that students make in educational activities that help them learn what they want to learn. This definition clarifies that engagement is more than just taking part; it is a multidimensional concept that includes mental, emotional, and behavioral involvement in the learning process.

Furthermore, Engagement with academic libraries helps students get help from the library and do well in school. For instance, Kuh and Gonyea (2023) were among the first to argue that academic libraries are essential for engaging students because they provide access to learning resources, study spaces, and technology that help students learn on their own or with others.

Consequently, Their results show that using the library is associated with better academic performance and greater involvement. Similarly, Ferguson and Cunningham (2019) found that when students and faculty have positive feelings toward librarians and library services, they are more likely to work together and become involved. They stated that students are much more likely to ask for help and take part in library-based learning activities when they see librarians as approachable academic partners.

Moreover, New studies continue to add to this picture. For example, MacMillan and Edwards (2025) examined ways to measure and improve students' involvement in academic libraries. Their research showed that feedback systems, active learning programs, and interactive library orientations make it much easier for students to connect with library spaces and resources. Thus, These strategies ensure engagement is not only measurable but also long-lasting. In addition, They also stress that the library is a place to learn, not just a place to store information.

Furthermore, Motivation and learning strategies are closely linked to engagement. For instance, Zhao and Wang (2025) found that effective learning strategies in both online and flipped classrooms positively affect students' motivation, engagement, and performance in school. They noted that students are more likely to be engaged in environments that are interactive, technology-rich, and encourage students to take part and work independently. Similarly, Liu et al. (2024) found a strong connection between pre-service teachers' achievement motivation, professional identity, and learning engagement. Their research demonstrated that students are more engaged when they

are driven to do well and strongly relate to their academic goals. Consequently, This, in turn, improves their practical learning outcomes.

Overall, All of these studies show that student engagement is a complex concept influenced by factors such as library settings, lesson plans, motivation, and school support. Therefore, Academic libraries are important places to get people involved because they offer active learning opportunities, easy access to resources, and friendly interactions. Hence, Institutions can significantly improve students' motivation and overall academic success by making libraries that encourage participation, reflection, and teamwork.

Learning satisfaction

When students reach their learning goals and feel good about themselves, that is called learning satisfaction. According to Li et al. (2021), satisfaction boosts motivation by recognizing students' hard work and encouraging them to keep learning.

Similarly, According to Zhao and Wang (2025), students are more likely to stay engaged and do well in school if they are happy with their learning environment and the way they are taught.In addition, Liu et al. (2024) also found that learning satisfaction is a link between motivation and achievement. Thus, This shows that being emotionally satisfied with the learning process is what keeps students going in school for a long time. Indeed, A key indicator of how students feel about their overall learning experience is their satisfaction with what they are learning. Moreover, Self-efficacy and intrinsic motivation affect how students study and how satisfied they are with the process, according to Li et al. (2023). Therefore, If students are self-motivated and confident in

their skills, they are more likely to be happy with their learning environment and what they learn. This demonstrates that satisfaction is not just about materials or facilities; it also depends on students' motivation and belief in their ability to learn well.

Furthermore, Peng et al. (2022) added to this by stating that the learning environment, especially the library, has a significant impact on students' happiness and productivity. For instance, When students think the library is well-organized and comfortable, they are more motivated to learn and happier with the process. Similarly, Viji and Balasubramanian (2020) also noted that students are happier with library services and resources when they are easy to access, updated frequently, and meet their academic needs.

In addition, Defrain et al. (2022) also found that libraries that encourage collaboration and have up-to-date facilities create spaces that are more interesting and flexible, thereby making students happier. Similarly, Mailu et al. (2023) also noted that learning satisfaction is not just about how well you do in school, but also about the overall quality of your learning experience. Consequently, Students are more enthusiastic, motivated, and persistent in their studies when they feel supported by the learning environment. Overall, These studies show that both internal and external factors cause learning satisfaction. Indeed, Internal factors include things like motivation and self-efficacy, whereas external factors include the quality of learning spaces and the resources institutions offer.

Theoretical Framework

This research is based on several learning and environmental design theories that

explain how a library's design affects students' motivation and learning interest.

Affordance Theory, introduced by psychologist James J. Gibson (1979), explains how the physical environment provides opportunities or “affordances” for action. In simpler terms, it shows how people interact with their surroundings based on what the environment allows them to do. In learning environments, this theory helps describe how features such as layout, lighting, acoustics, and furniture can influence students’ engagement, comfort, and motivation. According to Young and Cleveland (2022), educational spaces are most effective when their design affords meaningful actions like collaboration, focus, and creativity by aligning physical features with users’ needs.

In the context of libraries, Sadler and Given (2019) applied Affordance Theory to explain how students perceive and use library spaces. Their findings show that design elements such as open layouts, flexible seating, and quiet study zones afford various study behaviors and promote user satisfaction. Similarly, well-designed lighting, proper acoustics, ergonomic furniture, accessible technology, and the integration of green spaces afford comfort, focus, and psychological well-being. These environmental affordances make students more likely to stay motivated and engaged in their academic tasks.

The Self-Determination Theory, developed by Edward L. Deci and Richard M. Ryan in 1985, also guides this study. According to this theory, people are more intrinsically motivated when they feel autonomous, competent, and connected. A well-thought-out library encourages these things by providing areas for studying alone (autonomy), opportunities to improve skills and easy access to learning materials (competence), and opportunities to work together and socialize (relatedness).

Affordance Theory complements Self-Determination Theory (Deci & Ryan, 1985) by providing an environmental perspective on motivation. While Self-Determination Theory explains the internal needs that drive motivation autonomy, competence, and relatedness Affordance Theory explains how the library's physical design can help fulfill those needs. When a library affords choice, comfort, connection, and creativity through its design, students experience greater satisfaction and higher learning motivation. Therefore, Affordance Theory serves as a strong framework for understanding how library design shapes students' motivation to learn.

Schematic Diagram

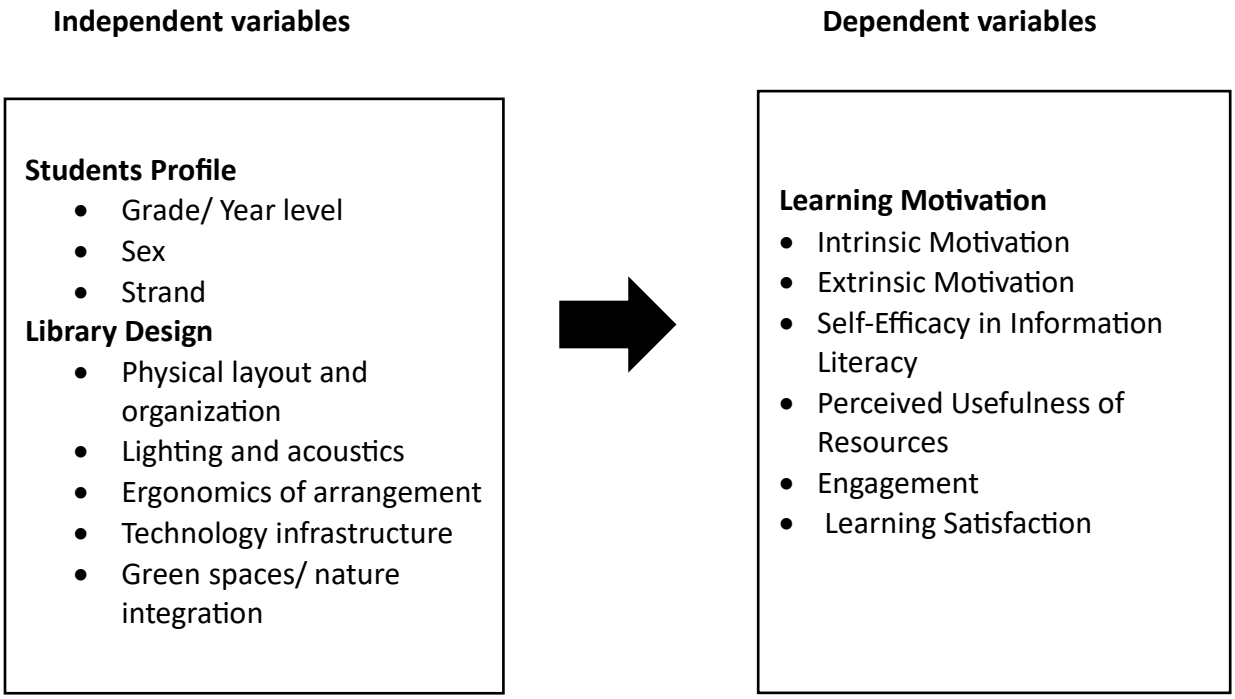


Figure 1. The figure shows the Relationship Between Students' Profile, Library Design, and Learning Motivation.

Statement of the Problem

The study aimed to determine the influence of library design on students' learning motivation at Saint Michael College of Caraga. The study aimed answers to the following questions:

1. What is the demographic profile of the respondents in terms of:

    1.1 Grade/Year and Level;

    1.2 Sex; and

    1.3 Strand?

2. What is the perceptions of the elementary, junior high, and senior high school students on the library design dimensions in terms of:

    2.2 Physical Layout and Organization;

    2.2 Lighting and Acoustics

    2.3 Ergonomics of Arrangement;

    2.4. Technology Infrastructure; and

    2.5. Green Spaces/Nature Integration?

3. What is the level of students' learning motivation in terms of:

    3.1 Intrinsic Motivation

    3.2 Extrinsic Motivation

    3.3 Self-efficacy in Information Literacy

    3.4 Perceived Usefulness of Resources

    3.5 Students Engagement

    3.6 Learning Satisfaction

4. Is there a significant relationship between the students' demographic profile and their learning motivation?

5. Is there a significant relationship between library design and students learning motivation?

6. Is there a significant relationship between the students' demographic profile and their level of perception of the library design?

Hypotheses of the Study

This part of the study is designed to identify and test the possible relationships among the variables of the research, particularly the students' demographic profile, library design, and learning motivation. The following hypotheses were formulated to determine whether significant relationships exist among these variables.

HO_1: There is no significant relationship between students' demographic profile (sex, grade level, and academic strand) and their learning motivation.

HO_2: There is no significant relationship between library design and students' learning motivation.

HO_3: There is no significant relationship between the students' demographic profile (sex, grade level, and academic strand) and their level of satisfaction with the library design.

Scope and Limitations

This study looks at how the Basic Education libraries are set up and what that means for learning motivation. Five main factors affect how well something works: the layout and design of the space; the lighting and sound; how comfortable the spaces are to be in; the technology used; and how well green spaces or nature are integrated. The study's goal is to determine how these design elements affect students' desire to use the library, their effective use of resources, and their participation in school activities.

For the School Year 2025–2026, the study will be limited to Basic Education students of Saint Michael College of Caraga. Specifically, only pupils from Grades 4 to 6 and students from Junior and Senior High School will serve as respondents. These grade levels are selected because learners at this stage possess the necessary comprehension and cognitive abilities to understand and accurately respond to survey questions. They also demonstrate greater independence in using the library for reading and academic tasks, making their insights vital to understanding the relationship between library design and learning motivation.

Younger pupils may find it difficult to articulate their perceptions, while college students function within different learning environments beyond the scope of this research. The survey cannot be accessed by teachers, librarians, or college students. The study focuses only on what students think and what motivates them. It does not measure their actual school performance. It also does not look at factors outside the library, such as family situation, personal study habits, or influence from friends. This study does not

examine external factors such as family background, personal study habits, or peer pressure.

Definition of Terms

Engagement. This refers to the level of students' active involvement in library use as measured by their self-reported frequency of participation in study activities, attention while using library spaces, and involvement in academic tasks within the library, as indicated in the engagement-related items of the survey questionnaire.

Ergonomics of Arrangement. This refers to students' perceived comfort, functionality, and ease of use of library furniture, seating, and spatial layout, as measured by their ratings on survey items related to seating comfort, workspace adequacy, and physical convenience within the library.

Extrinsic Motivation. This refers to the extent to which students are motivated to use the library due to external factors such as grades, academic requirements, teacher encouragement, or recognition, as measured by their responses to questionnaire items assessing external motivational influences.

Green Spaces/Nature Integration. This refers to the presence and perceived impact of natural elements in the library (e.g., plants, open areas, natural lighting, and outdoor views), measured through students' ratings of how these elements contribute to relaxation, focus, and comfort while studying.

Intrinsic Motivation. This refers to students' internal drive to use the library for personal interest, enjoyment, and self-improvement, measured by their agreement with survey statements indicating voluntary library use without reliance on external rewards.

Learning Satisfaction. This refers to students' overall level of satisfaction with their learning experiences in the library, measured through survey items assessing contentment, perceived effectiveness, and positive learning experiences while using library facilities.

Lighting and Acoustics. This refers to students' perception of the adequacy of lighting and sound conditions in the library, including brightness, availability of natural light, noise control, and sound comfort, as measured by their responses to specific questionnaire items on environmental quality.

Perceived Usefulness of Resources. This refers to students' belief in the usefulness and relevance of library materials, facilities, and services in achieving academic goals, measured by their ratings of the availability, adequacy, and helpfulness of library resources.

Physical Layout and Organization. This refers to students' perception of how well library spaces, furniture, and materials are arranged and organized, measured by survey items related to accessibility, ease of navigation, seating arrangement, and logical placement of resources.

Self-Efficacy in Information Literacy. This refers to students' confidence in their ability to locate, evaluate, and use information effectively using library resources, measured through self-assessment items addressing information-searching skills and resource utilization.

Sex. This refers to the biological classification of the student respondents as male or female, as self-reported in the demographic section of the questionnaire, and used as a grouping variable for data analysis.

Strand. This refers to the academic track of the students (e.g., HUMSS, STEM, ABM, TVL), as indicated in the demographic section of the questionnaire, and used to compare perceptions of library design and learning motivation across groups.

Technology Infrastructure. This refers to the availability, accessibility, and reliability of technological facilities in the library, such as computers, Wi-Fi connectivity, charging ports, and digital resources, as measured by students' ratings of technology-related survey items.

CHAPTER 2

RESEARCH METHODOLOGY

This chapter discusses the study's research design, respondents, instruments, data-gathering procedure, and statistical techniques.

Research Design

This study adopts a descriptive-correlational research design to determine how library design is related to students' learning motivation at Saint Michael College of Caraga (SMCC). This design is suitable because the study aims to describe existing conditions of the library environment and examine the relationship between library design features and students' motivation without manipulating any variables.

The descriptive component of the design allows the researcher to gather and present students' perceptions of the current library design, including physical layout, lighting, acoustics, ergonomic arrangements, technology facilities, and nature-inspired spaces. This approach provides a clear and systematic description of how students experience and evaluate the library environment, as well as their levels of learning motivation.

Meanwhile, the correlational component is appropriate for identifying the degree and direction of the relationship between library design elements and students' learning motivation. Since the study seeks to determine whether variations in library design are associated with differences in students' motivation, correlational analysis enables the examination of these relationships under natural conditions. No experimental

intervention is required, making this design practical and ethical within an educational setting.

Overall, the descriptive-correlational research design is well-suited to this study because it provides both a detailed description of the library environment and a statistical basis for understanding how library design is related to students' learning motivation. This integrated approach ensures a comprehensive analysis of the problem within the real context of SMCC.

Research Locale

The study will be conducted at Saint Michael College of Caraga (SMCC) is a private educational institution in Nasipit, Agusan del Norte, offering programs from Elementary to Senior High School. The college has two main campuses: the Main Campus in Barangay 4, which houses the Junior High School, and the Triangulo Campus, which accommodates the Elementary and Senior High School departments. Both campuses have a substantial student population, well-organized facilities, and a supportive learning environment, making them ideal for studying the impact of library design on students' motivation and learning outcomes. The campuses' accessible locations allow for smooth data collection and observation of student interactions with library facilities.

School A. Located along Atupan Street in Barangay 4, (Main Campus) houses the Junior High School. (Lat. 8.7450° N, Long. 125.5280° E).

School B. Accommodates the Elementary and Senior High School departments in Barangay Triangulo (Lat. 8.7485° N, Long. 125.5342° E).

Both campuses are marked by a substantial student population and provide a

secure, supportive learning environment. Their strategic locations and accessibility facilitate efficient data collection and observational procedures.

Research Respondents

The respondents of this study will be drawn from the Basic Education Department of Saint Michael College of Caraga (SMCC), located in Nasipit and Triangulo, Agusan del Norte. These respondents will include students from Elementary Grades Four to Six, Junior High School, and Senior High School levels.

Grade 4 will be selected as the entry level of the study because learners at this stage are expected to possess adequate reading skills and the ability to provide clear and independent responses to survey questionnaires. Together with Grades 5 and 6, Grade 4 learners will regularly utilize the school library, ensuring that the respondents are familiar with the library environment. This group of respondents is expected to provide meaningful insights into how library design will influence students' learning motivation and engagement.

The study will employ random sampling, wherein each student in the population will have an equal chance of being selected as a respondent. To determine the appropriate sample size, Slovin's Formula will be applied, as it is suitable when the total population is known.

Table 1. Research Respondents

Educational Level	Male	Female	N	N
Elementary (Grades 4–6)	13	13	52	26
Junior High School (Grades 7–10)	139	142	940	281
Senior High School (Grades 11–12)	109	130	591	239
Total	**261**	**285**	**1583**	**546**

As shown in Table 1, a total of 546 respondents, representing approximately 50% of the total student population of 1,583**,** will participate in the study. This distribution will ensure that the data to be collected will be well-balanced and representative across the elementary (Grades 4–6), junior high school, and senior high school levels.

Research Instrument

The primary quantitative instrument employed in this study is a researcher-made questionnaire, designed to gather data from students in the elementary, junior high, and senior high levels of the Basic Education department at Saint Michael College of Caraga. The questionnaire is divided into three major sections to ensure comprehensive data collection. Part I focuses on the demographic profile of respondents, including age, sex, and educational level. Part II examines the study's independent variables, which pertain to key aspects of library design such as Physical Layout and Organization, Lighting and Acoustics, Ergonomics of Arrangements, Technology Infrastructure, and Green Spaces/Nature Integration. Part III explores the dependent variables related to student motivation and learning outcomes, including Learning Motivation, Intrinsic Motivation, Extrinsic Motivation, Self-efficacy in Information Literacy, Perceived Usefulness of Resources, Student Engagement, and Learning Satisfaction.

The questionnaire employs a five-point Likert scale to measure respondents' level of agreement with each statement. The scale points are: 1 – Strongly Disagree, meaning I do not agree at all; this is not true for me; 2 – Disagree, meaning I mostly do not agree; it is usually not true for me; 3 – Agree, meaning I mostly agree; it is true for me most of the time; and 4 – Strongly Agree, meaning I completely agree; this is very true for me. This scale was chosen for its clarity and simplicity, making it appropriate for students across all levels while allowing sufficient range for statistical analysis.

The questionnaire consists of a total of 85 items, distributed across Part I demographic profile: 5 items, Part II library design elements: 50 items, and Part III learning motivation and outcomes: 30 items. To ensure clarity, relevance, and alignment with the study's objectives, the instrument will undergo content validation by three professionals with expertise in education and research. A pilot test will also be conducted prior to full deployment to assess reliability and refine the instrument as needed.

Once finalized, the collected data will be encoded and analyzed using appropriate statistical tools. Descriptive statistics, such as frequencies and percentages, will summarize the participants' demographic profile, while weighted means will be used to assess perceptions of library design and levels of learning motivation. If necessary, inferential statistics, such as Pearson's r, will determine the presence and strength of relationships between the independent variables library design elements and dependent variables learning motivation factors, with the final selection of statistical tools guided by consultation with the research adviser to ensure methodological rigor.

Research Ethical Standards

ETHICAL CONSIDERATIONS

A. Intellectual Property Rights (IPR)

The study does not create any new technology, process, or product that requires patent protection. It focuses on how library design affects students' learning motivation. All sources and authors are properly cited to respect intellectual property rights. Questionnaires are used only for academic purposes and not for profit. No logos, brands, or trademarks will be involved, and the name Saint Michael College of Caraga will be mentioned only for academic use.

B. Informed Consent

All respondents will be informed about the study's purpose, procedures, roles, and the possible risks and benefits of participation. Participation will be voluntary, and respondents may withdraw at any time without penalty. Confidentiality will be strictly observed, and respondents' identities will not be disclosed. An informed consent form will be provided and signed after respondents fully understand the study.

For respondents who are minors, a separate parental informed consent form will be required. Parents or legal guardians will be given a distinct document explaining the study and must provide written permission before their child's participation. This document will be attached as an appendix.

The study will involve human respondents only and will follow ethical standards for human research. The informed consent and parental informed consent processes will ensure the protection of the rights, welfare, and dignity of all respondents.

C. Data Privacy and Confidentiality

Data will be collected through voluntary survey questionnaires without requiring any personally identifiable information. All responses will be kept strictly confidential and used only for academic purposes.

The data will be securely stored in password-protected digital files or locked storage accessible only to the researcher. Upon completion and approval of the study, printed materials will be shredded and electronic files will be permanently deleted. Respondents will be assured that their information will not be shared with any third party and that their identities will remain anonymous at all times.

D. Voluntary Participation

All participants will join the study voluntarily after understanding its purpose. They will be informed that they may withdraw at any time without consequence, and their responses will remain confidential. Participation or withdrawal will not affect them in any way. No animals will be involved in this study.

E. Minimization of Harm

This study will not pose any physical, psychological, or emotional risk to participants because it only involves answering questions about their experiences and perceptions of the library. Nonetheless, the researchers will prioritize the safety and well-being of all students throughout the data-gathering process. If any student shows signs of physical discomfort such as dizziness, fever, or illness they will be asked to stop answering the questionnaire and will be immediately referred to the school clinic for proper care. If a participant feels uncomfortable or distressed at any point, they may skip any question

or withdraw from the study without any penalty. Throughout the process, the researchers will remain attentive and respectful to ensure a safe and supportive research environment for all participants.

No animals or living organisms will be involved, and all procedures will follow ethical standards for human subject research. The study will prioritize the safety, well-being, and rights of all participants, ensuring that the research environment remains respectful, secure, and free from harm.

F. Beneficence

The study will benefit educators, librarians, and students by providing insights into how library design supports learning motivation. Findings may help improve study spaces, encourage engagement, and enhance academic performance. Participants may also request access to the study results to understand how their feedback contributed to the findings.

G. Justice and Fair Participation Selection

All respondents will be treated fairly, regardless of gender, age, or background, and participation will be limited to students who will meet the study's criteria. Respondents will be selected using appropriate sampling procedures to ensure fairness, and the survey will be designed to ensure accuracy and reliability. Potential biases and limitations, such as those arising from self-reported data, will be acknowledged and will be considered to ensure the ethical and responsible presentation of the findings.

H. Data Integrity and Accuracy

Standardized methods will be used to ensure accuracy and reliability in data

collection and analysis throughout the study. Potential biases, errors, and limitations will be documented and considered to maintain the validity and integrity of the research results.

I. Transparency in Reporting

The researcher will report all findings truthfully and without manipulation. All materials, sources, and tools adapted from other works will be properly cited to ensure honesty and transparency. This study will utilize AI tools to support data organization and analysis. The researcher declares that there are no conflicts of interest related to this study.

J. Use of Patented or Copyrighted Materials

The study will not use any patented technologies or copyrighted materials that require permission. All sources and references will be cited appropriately to maintain academic integrity.

K. Ethical Considerations for Animal and Human Trials

Participants will be informed of the minimal risks and potential benefits of the study and assured that no physical, psychological, or emotional harm is expected. Support services will be available in case of any emergencies. No animals will be involved, and the study will follow ethical standards to protect the well-being, rights, and dignity of all human respondents.

L. Responsible Use of AI and Related Technologies

AI tools will not use for data collection or analysis. They are used only for technical

purposes such as formatting or grammar checking. All data will be manually collected and analyzed by the researcher to maintain integrity and accuracy.

M. Ethical Approval

The researcher will seek ethical approval from the Saint Michael College of Caraga Research Ethics Committee (SMCC-REC) before data collection begins. This will ensure that the study will follow all ethical standards for research involving human participants.

Data Gathering Procedure

This study will be conducted with the prior permission of the school authorities, including the Dean and Principals, to carry out the research titled “Library Design: Its Implication on Learning Motivation.” Upon approval, the researchers will personally deliver transmittal letters, informed consent forms, and questionnaires to the selected respondents from the Elementary, Junior High, and Senior High School departments. For minor respondents, parental consent will also be obtained before participation. Only approved students will be included as respondents.

The respondents will be informed about the purpose, goals, procedures, risks, and benefits of the study. They will be assured that participation is voluntary, and all responses will be kept confidential and used solely for academic purposes. Sufficient time will be provided to answer each questionnaire item honestly and carefully. The questionnaire will consist of three parts: (1) Basic Information, (2) Closed-ended Questions, and (3) Likert Scale Questions. After completion, the researchers will personally collect the questionnaires, verify them for completeness and accuracy, and prepare the data for analysis.

To ensure an organized and smooth data-gathering process, the researchers will seek assistance from the School Principal and the Librarian assigned to orient the respondents and facilitate proper administration of the questionnaires.

Letter of Approval. Before conducting the study, the researchers will secure a letter of approval from the school authorities, including the Dean and School Principals. This formal permission ensures that the research is officially recognized and allowed within the school.

Informed Consent. The researchers will inform the selected respondents about the purpose of the study, its goals, procedures, potential risks, and benefits**.** Respondents will be made aware that participation is voluntary and that their responses will remain confidential and used solely for academic purposes. For minor respondents, parental consent will also be obtained. Only those who give consent will be included in the study.

Distribution of the Questionnaire. After approval and consent, the researchers will personally deliver the questionnaires to the selected respondents from the Elementary, Junior High, and Senior High School departments. Respondents will be given instructions on how to complete the questionnaire and sufficient time to answer all items honestly and carefully**.**

Gathering of Data. Once the respondents have completed the questionnaires, the researchers will personally collect them to ensure that all responses are accounted for. The completed questionnaires will then be checked for completeness and accuracy to ensure the data is ready for analysis.

Data Analysis Plan

In presenting, interpreting, and analyzing the data gathered by the researchers,

the following statistical measures will be used to treat the data:

Frequency and Percentage. These will be used to describe the distribution of respondents' demographic information and their responses to closed-ended questions.

Mean. This will be used to determine respondents' level of agreement with the statements in the Likert-scale section, reflecting their perceptions of the library's physical design and its impact on their learning motivation.

Quantitative analysis. Will employ closed-ended questions to collect numerical data that can be measured and statistically analyzed. This approach allows the researchers to objectively determine the relationship between library design and students' learning motivation.

The results will be presented in tabular form and interpreted based on the Statement of the Problem of the study.

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CURRICULUM VITAE



NAME: Ethel Joy M. Longakit
ADDRESS: P-3 Afga, Sibagat, Agusan del Sur
EMAIL: etheljoy_longakit@smccnasipit.edu.ph

Personal Information

Sex : Female
Date of birth : October 23, 2004
Height : 5'5
Weight : 50klg.
Civil Status : Single

Educational Attainment

Elementary : Afga Central Elementary School
Secondary : Father Saturnino Urios College of Sibagat Inc.
Tertiary : Saint Michael College of Caraga

Membership/Affiliations

ORGANIZATION	POSITIONS HELD
Local Student Government	BEED Representative

CURRICULUM VITAE



NAME: Niña Beatrice L. Capuyan
ADDRESS: District-2 Punta, Nasipit, Agusan del Norte
EMAIL: nina_capuyan@smccnasipit.edu.ph

Personal Information

Sex : Female
Date of birth : January 17,2005
Height :4'9
Weight :55
Civil Status : Single

Educational Attainment

Elementary : Punta Elementary School
Secondary : Northwestern Agusan Colleges
Tertiary : Saint Michael College of Caraga

Membership/Affiliations

ORGANIZATION	POSITIONS HELD

CURRICULUM VITAE



Name: Arvie Jane F. Celis
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Personal Information

Sex : Female
Date of birth : July 25,2005
Height :5'3
Weight :55
Civil Status : Single

Educational Attainment

Elementary : Bangonay Central Elementary School
Secondary : Buenavista National High School
Tertiary : Saint Michael College of Caraga

Membership/Affiliations

ORGANIZATION	POSITIONS HELD

CURRICULUM VITAE



Name: Charish V. Claro
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Personal Information

Sex : Female
Date of birth : June 12,2005
Height : 5'6
Weight :52
Civil Status : Single

Educational Attainment

Elementary : Lingayao Elementary School
Secondary : Lingayao National High School
Tertiary : Saint Michael College of Caraga

Membership/Affiliations

ORGANIZATION	POSITIONS HELD

CURRICULUM VITAE



Name: Ana Marie L. Reducto
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EMAIL: anamarie_reducto@smccnasipit.edu.ph

Personal Information

Sex : Female
Date of birth : October 18, 2004
Height :5'1
Weight :42
Civil Status : Single

Educational Attainment

Elementary : Afga Central Elementary School
Secondary : Afga National high school
Tertiary : Saint Michael College of Caraga

Membership/Affiliations

ORGANIZATION	POSITIONS HELD

APPENDIX A-1

A - Letter of Approval

TRANSMITTAL LETTER

October 28, 2025

Rene A. Japitana, EdD-CAR
Principal SMCC, High School
Brgy. 4 Atupan, Nasipit, Agusan del Norte

Dear Sir Japitana,

Good day!

We are a group of 3rd year Bachelor of Elementary Education students from Saint Michael College of Caraga currently conducting our research titled "Library Design: Its Implication on Learning Motivation".

This study specifically aims to determine the problems and challenges that students experience in relation to the library's physical design, arrangement, and facilities, and how these factors affect their motivation to learn. As part of our data-gathering process, we would like to request your permission to conduct interviews with selected students inside the campus. We will ensure that the interviews will be done quietly, in accordance with library policies, and without disturbing other users.

We assure you that all gathered information will be used solely for academic purposes and treated with confidentiality.

We are hoping for your kind approval. Thank you very much for your time and support.

Respectfully yours,

ETHEL JOY M. LONGAKIT
Research Leader

Noted by:

RHEAN L. SANCHEZ, EdD
Research Adviser

Approved by:

RENE A. JAPITANA, EDD-CA
Principal

SAINT MICHAEL COLLEGE OF CARAGA
RESEARCH & INSTRUCTIONAL INNOVATION DEPARTMENT
www.smccnasipit.edu.ph

TRANSMITTAL LETTER

October 29, 2025

Berna Jessa M. Ganzan, OIC
SMCC- Elementary Principal
Brgy. Triangulo, Naisipit, Agusan del Norte

Dear Ma`am Layong,

Good day!

We are a group of 3rd year Bachelor of Elementary Education students from Saint Michael College of Caraga currently conducting our research titled "Library Design: Its Implication on Learning Motivation".

This study specifically aims to determine the problems and challenges that students experience in relation to the library's physical design, arrangement, and facilities, and how these factors affect their motivation to learn. As part of our data-gathering process, we would like to request your permission to conduct interviews with selected students inside the campus. We will ensure that the interviews will be done quietly, in accordance with library policies, and without disturbing other users.

We assure you that all gathered information will be used solely for academic purposes and treated with confidentiality.

We are hoping for your kind approval. Thank you very much for your time and support.

Respectfully yours,

ETHEL JOY M. LONGAKIT
Research Leader

Noted by:

RHEAN L. SANCHEZ, EDD
Research Adviser

Approved by:

Berna Jessa M. Ganzan, OIC
Elementary Principal

APPENDIX B-1

B. Research Instruments

SURVEY QUESTIONNAIRE

Dear Respondent,

This questionnaire aims to gather information about how the design of the library (layout, lighting, furniture, atmosphere, etc.) affects students' motivation to learn. Your honest responses will help us understand how the library environment influences your interest, engagement, comfort, and satisfaction while studying or using its facilities.

The data you provide will be used solely for academic research purposes. Your participation is voluntary, and all your answers will be treated with strict confidentiality. There are no right or wrong answers; we simply ask you to respond based on your personal experiences and perceptions.

Thank you for taking the time to participate in this study. Your input is valuable and greatly appreciated.

PART I – DEMOGRAPHIC PROFILE OF RESPONDENTS

Directions: Please put a check **(✓)** mark in the box. Choose only one answer except for number 5.

Name (optional): _____

1. sex:
   ☐ male
   ☐ female

2. [single choice] You are:
   ☐ elementary level
   ☐ junior high school level
   ☐ senior high level

3. On average, how often do you go to the library this semester:
   ☐ every day
   ☐ weekly

☐ monthly
☐ quarterly
☐ basically
☐ others
Please specify:

4. On average, the time you spend in the library each time is about:
   ☐ less than 30 min
   ☐ 30 min–1 h
   ☐ 1–3 h
   ☐ more than 3 h
   ☐ others
   Please specify:

5. Your main purpose of going to the library is to/for:
   ☐ self-study
   ☐ complete group work
   ☐ borrow and return books
   ☐ speech practice
   ☐ use electronic resources in the library for learning
   ☐ conduct social activities such as communication and interaction with others
   ☐ entertainment (watching movies, listening to music, playing games, etc.)
   ☐ others
   Please specify:

Instruction:

Please read each statement carefully and put a check mark **(✓)** on the space that best represents your level of agreement. There are no right or wrong answers. Your honest response will help in gathering accurate data for this study. Rest assured that all answers will be treated with utmost confidentiality and used only for research purposes.

(*Palihug basaha ang matag pamahayag ug ibutang ang check mark* **(✓)** *sa kahon nga nagsulti sa imong tinuod nga pagsabot. Walay sakto o sayop nga tubag. Ang imong tapat nga tubag makatabang sa pagkuha ug husto nga datos para niining pagtuon. Sigurado nga tanang tubag bantayan ug taguon ug gamiton ra para sa research).*

Scale:
1– Strongly Disagree: I do not agree at all. This is not true for me.
2–Disagree: I mostly do not agree. It is usually not true for me.
3–Agree: I mostly agree. It is true for me most of the time.
4–Strongly Agree: I completely agree. This is very true for me.

A. Physical Layout and Organization	**Strongly Disagree (1)**	**Disagree (2)**	**Strongly Agree (3)**	**Agree (4)**
1. The library's layout allows easy movement between shelves and tables. Ang paagi sa pagbutang sa libraryo nagpasabot nga sayon ra molihok gikan sa estante padulong sa lamesa.				
2. The arrangement of books and sections helps students find materials quickly. Ang pagbutang sa mga libro ug lugar sa libraryo makatabang nga sayon ra makakita ug makakuha sa gusto nga libro.				
3. The study areas are properly separated for group and individual study. Ang mga lugar sa pagtuon klaro ang bulag para sa grupo ug para sa tagsa-tagsa nga pagtuon.				
4. The design of the library encourages students to stay and study longer. Ang disenyo sa library nagdasig nako nga magpabilin ug magtuon ug dugay.				
5. The library space is well-organized and visually appealing. Ang espasyo sa library kay maayong gi-organisar ug nindot tan-awon.				
6. There is enough space between tables to maintain comfort and privacy. Ado'y igo nga luna tali sa mga lamesa aron mapreserba ang kahupayan ug privacy.				
7. The signage and directions inside the library are clear and easy to follow. Ang mga timaan ug direksyon sa library kay klaro ug sayon sundon.				
8. The library layout supports smooth traffic flow even during busy hours. Ang layout sa library nagsuporta sa hapsay nga paglihok bisan kon daghang tawo.				
9. The overall structure of the library promotes effective studying. Ang kinatibuk-ang porma sa library nagtabang aron mas epektibo ang pagtuon.				
10. The library layout makes the environment feel open and welcoming.				

Ang layout sa library naghimo nga ang lugar mura'g abli ug makapahimuot.				
B. Lighting and Acoustics				
1. The lighting inside the library is bright enough for reading and studying. Ang kahayag sulod sa library igo kaayo para sa pagbasa ug pagtuon.				
2. The natural light enters the library and makes it feel lively and comfortable. Ang natural nga kahayag misulod sa library ug naghimo niini nga mas komportable ug buhi ang dating.				
3. The artificial lights are properly placed to avoid shadows on reading tables. Ang mga suga nga artipisyal gibutang sa husto aron malikayan ang mga landong sa lamesa.				
4. The brightness of the library does not strain my eyes. Ang kasanag sa library dili makapahago sa akong mata.				
5. The library maintains a quiet environment for concentration. Ang library nagpabilin nga hilum aron makapokus sa pagtuon.				
6. The noise from outside does not easily enter the library. Ang kasaba gikan sa gawas dili dali makasulod sa library.				
7. The soundproofing or wall design helps reduce noise. Ang soundproof nga disenyo sa bongbong makatabang nga maminusan ang kasaba.				
8. The proper lighting helps me stay alert while studying. Ang sakto nga kahayag makatabang nako nga magpabilin nga aktibo samtang nagtuon.				
9. The light temperature feels comfortable and not too harsh. Ang kahayag dili sakit sa mata ug komportable ra tan-awon.				
10. The overall lighting and sound environment motivate me to study longer.				

Ang kinatibuk-ang kahayag ug kalinaw sa lugar nagdasig nako nga magtuon ug dugay.)				
C. Ergonomics of Arrangements				
1. The chairs and tables in the library are comfortable for long study hours. Ang mga bangko ug lamesa sa library komportable gamiton bisan dugay magtuon.				
2. The furniture is appropriate for the students height and posture. Ang mga kasangkapan angay sa akong kahabogon ug posisyon sa lawas.				
3. The library provides enough seating for all students. Ang library naghatag igo nga lingkuranan para sa tanan nga estudyante.				
4. The tables are spacious enough for my study materials. Ang mga lamesa kay dako ug igo para sa akong mga gamit sa pagtuon.				
5. The seating arrangement supports both individual and group study. Ang paghan-ay sa lingkuranan maayong para sa solo ug panggrupo nga pagtuon.				
6. The furniture design reduces physical discomfort during study. Ang disenyo sa mga kasangkapan makatabang nga dili kapoy sa lawas.				
7. The spacing between chairs and tables prevents crowding. Ang gilay-on sa mga bangko ug lamesa makalikay sa kahuot.				
8. The furniture is well-maintained and stable. Ang mga kasangkapan sa library kay maayong kondisyon ug lig-on.				
9. The comfortable furniture helps student concentrate better. Ang komportableng lingkuranan makatabang nga makapokus ko sa pagtuon.				
10. The overall ergonomics of the library furniture increases students motivation to study.				

Ang hapsay nga disenyo sa kasangkapan sa library nagpadako sa akong gana sa pagtuon.				
D. Technology Infrastructure				
1. The library provides stable Wi-Fi access for studying and research. Ang library naghatag ug lig-on nga Wi-Fi para sa pagtuon ug research.				
2. The computers are available and functional for student use. Ado'y mga computer nga magamit ug maayo ang kondisyon.				
3. The library offers access to online journals and databases. Ang library naghatag ug access sa mga online journal ug database.				
4. The library has enough power outlets or charging stations for devices. Ang library adunay igo nga saksakan para sa pag-charge sa mga gamit.				
5. The use of technology makes studying easier and faster. Ang paggamit sa teknolohiya nagpasayon ug nagpadali sa pagtuon.				
6. The digital materials are easy to access through the library system. Dali ra nako ma-access ang mga digital nga materyales pinaagi sa system sa library.				
7. The library's technology resources support collaboration and group work. Ang teknolohiya sa library nagtabang sa pagtinabangay sa grupo.				
8. Technical support is available when needed. Ado'y mutabang sa teknikal nga problema kung kinahanglan.				
9. The combination of physical and digital resources enhances students' learning. Ang pagsagol sa physical ug digital nga gamit nakapabaskog sa akong pagtuon.				
10. The library's technology encourages me to visit and use its services more often.				

Ang teknolohiya sa library nagdasig nako nga mas kanunay gamiton kini.				
E. Green Spaces / Nature Integration				
1. The presence of plants makes the library environment more relaxing. Ang mga tanom sa library nakapahapsay sa lugar ug naghatag ug relaxing nga kahimtang.				
2. The library design includes natural light and ventilation. Ang disenyo sa library naglakip sa natural nga kahayag ug hangin.				
3. The use of natural colors and materials creates a calm study atmosphere. Ang paggamit sa natural nga kolor ug materyales naghatag ug malinawon nga atmospera.				
4. The windows provide a view of nature or open space. Ang mga bintana nagpakita sa tan-awon sa kinaiyahan o lapad nga lugar.				
5. The natural elements make the library more welcoming. Ang mga elemento sa kinaiyahan naghimo sa library nga mas makapahimuot.				
6. The presence of greenery helps reduce stress among students. Ang mga tanom makatabang nga maminusan ang akong kakapoy o stress.				
7. The natural lighting in the library helps students stay focused. Ang natural nga kahayag sa library makatabang nga magpokus ko sa pagtuon.				
8. The environment feels fresh and less confined because of natural elements. Tungod sa mga natural nga elemento, ang lugar bugnaw ug dili sapoton.				
9. The green spaces improve my mood while studying. Ang mga tanom ug greenery makapabati ug maayong pagbati samtang nagtuon.				
10. The integration of nature into the library design motivates students to stay longer.				

Ang presensya sa kinaiyahan sa disenyo sa library nagdasig nako nga magpabilin ug dugay.				
F. Intrinsic Motivation				
1. I use the library because exploring information increases my personal interest and enjoyment in learning. Gamiton nako ang librarya kay ang pag-explore sa impormasyon makapataas sa akong interes ug kalipay sa pagkat-on.				
2. I use the library because it helps me learn my lessons well. Gamiton nako ang librarya kay makatabang kini nako nga makat-on sa akong mga leksyon og tarong.				
3. I continue using the library even without external rewards because gaining knowledge is fulfilling to me. *Padayon ko nga gamiton ang librarya bisan walay ganti kay ang pagkat-on makahatag nako og kalipay.*				
4. I frequently visit the library because it supports my personal educational goals and lifelong learning habits. *Permi ko moadto sa librarya kay kini mosuporta sa akong personal nga edukasyonal nga mga tumong ug sa akong tibuok kinabuhi nga pagkat-on.*				
5. I enjoy using the library to read journals, borrow reference books for my lessons, and do research for school projects. These activities help me understand my subjects better and give me a clear sense of progress in my learning. *Ganahan ko mogamit sa library para mobasa og journals, manguha og reference books para sa akong lessons, ug mo-research para sa school projects. Kini nga mga kalihokan makatabang nako masabtan ang akong mga subjects og makahatag nako og klarong pagbati nga nag-uswag ko sa akong pagtuon.*				
G. Extrinsic Motivation				

1. I use the library when it is required for completing assignments or academic projects. *Gamiton nako ang librarya kung kini gikinahanglan aron makompleto ang mga assignments o academic projects.*				
2. I am motivated to go to the library because it helps me achieve higher grades and academic recognition. Na-motivate ko moadto sa librarya kay kini makatabang nako makab-ot ang taas nga grado ug akademikong pagkaila.				
3. I rely on library resources because professors and institutional requirements encourage me to do so. Nagsalig ko sa resources sa librarya kay ang mga propesor ug requirements sa institusyon nag-awhag nako niini.				
4. I visit the library to fulfill performance expectations and meet course requirements. Moadto ko sa librarya aron matuman ang mga performance expectations ug mga requirement sa kurso.				
5. I am encouraged to use the library when I believe it will contribute to my academic performance evaluation. Na-awhag ko sa paggamit sa librarya kung nagtuo ko nga kini makatabang sa akong academic performance evaluation.				
G. Self-Efficacy in Information Literacy				
1. I am confident that I can successfully locate relevant academic information in the library. Kompiyansa ko nga makakaplag ko og angay nga akademikong impormasyon sa librarya.				
2. I believe I can evaluate whether information from the library is credible for academic use. Nagtuo ko nga makasusi ko kung ang impormasyon sa librarya kasaligan ba para sa akademikong gamit.				

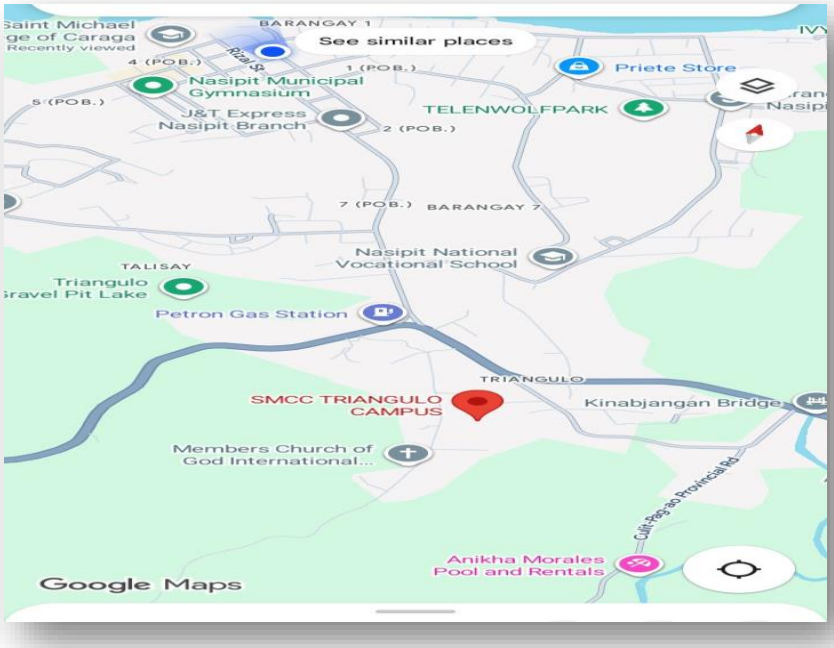
3. I feel capable of using library databases to complete my research tasks efficiently. Nabati nako nga kaya nako gamiton ang library databases aron epektibong makompleto ang akong research tasks.				
4. I am confident that I can use both digital and print resources to meet academic requirements. Kompiyansa ko nga magamit nako ang digital ug print resources aron matuman ang akademikong requirements.				
5. I believe my ability to find and use information improves my motivation to continue learning. Nagtuo ko nga ang akong abilidad sa pagpangita ug paggamit sa impormasyon makapauswag sa akong motibasyon sa pagpadayon sa pagkat-on.)				
H. Perceived Usefulness of Resources				
1. I believe that the resources in the library contribute directly to my academic success. Nagtuo ko nga ang mga resources sa librarya direkta nga nakatabang sa akong akademikong kalampusan.				
2. I find the library resources valuable because they help improve my learning performance and outcomes. Gitan-aw nako nga bililhon ang mga resources sa librarya kay kini makatabang sa pagpaayo sa akong performance ug resulta sa pagkat-on.				
3. I use the library regularly because the materials are relevant to my coursework. Permi ko gamiton ang librarya kay ang mga materyales angay sa akong coursework.				
4. I consider digital library platforms useful for enhancing my academic understanding. Gitan-aw nako nga mapuslanon ang digital library platforms para mapalambo ang akong akademikong pagsabot.				
5. I believe that updated and accessible resources in the library motivate me to continue studying.				

Nagtuo ko nga ang updated ug dali ma-access nga mga resources sa librarya makamotivate nako sa pagpadayon sa pagtuon.				
I. Engagement				
1. I actively participate in learning activities when using collaborative areas in the library. Aktibo ko nga moapil sa mga kalihokan sa pagkat-on sa paggamit sa collaborative areas sa librarya.				
2. I feel more involved in my studies when I use technology-integrated library spaces. Mas apil ko sa akong pagtuon kung gamiton nako ang technology-integrated library spaces.				
3. I invest more time and effort in studying when I am in the library. Gahin nako ang dugang oras ug kusog sa pagtuon kung naa ko sa librarya.				
4. I use the library as a space to interact and share learning ideas with others. Gamiton nako ang librarya ingon nga lugar sa pag-interact ug pag-share sa mga ideya sa pagkat-on sa uban.				
5. I feel more engaged when the library offers active learning programs and support services. Mas maengganyo ko kung ang librarya nagtanyag og mga aktibong programa sa pagkat-on ug mga serbisyo nga mosuporta sa mga estudyante.				
J. Learning Satisfaction				
1. I feel satisfied with my learning experience when the library provides a supportive academic environment. Nalipay ko sa akong learning experience kung ang librarya nagtanyag og supportive academic environment.				
2. I am emotionally fulfilled when library resources help me achieve my learning goals. Na-feel nako nga emotionally fulfilled kung ang mga resources sa librarya makatabang nako makab-ot ang akong learning goals.				

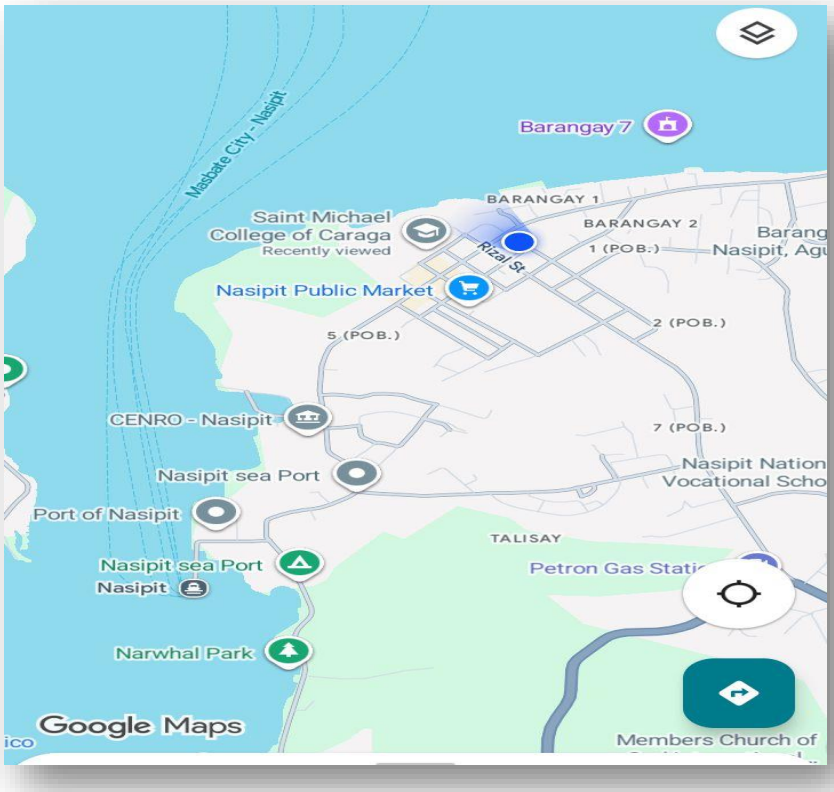
3. I feel more content with my academic progress when using well-designed library facilities. Mas kontento ko sa akong akademikong progreso kung gamiton nako ang maayong disenyo sa library facilities.				
4. I am satisfied when library services make my learning process easier and more effective. Nalipay ko kung ang library services nagpahapsay ug nagpalambo sa akong learning process.				
5. I feel that using the library positively contributes to my overall satisfaction as a student. Nabati nako nga ang paggamit sa librarya positibo nga nakatampo sa akong kinatibuk-ang kasatisfaksyon ingon nga estudyante.				

APPENDIX C

C - Map of the Research Locale



Coordinates: 8.7485° N, 125.5342° E



Coordinates: 8.7450° N, 125.5280° E

APPENDIX D

D. Documentation



Our group gathered to start drafting Chapter 1 of our study. We discussed the research title, the background of the study, and the statement of the problem. Everyone shared their ideas and built a strong foundation for our paper.



We presented our initial draft to our research adviser for review and guidance. The adviser provided valuable feedback and highlighted areas that needed improvement. This consultation helped us understand how to strengthen and refine our study.



We began revising our work based on the adviser's comments. Together, we corrected grammatical errors, improved coherence, and enhanced the overall content of our study.



Our group gathered to discuss the comments that needed revision. We also began revising immediately to make the process more efficient.